

2EI4 – Electronic Devices and Circuits I

Project 2 Research – Voltage Controlled Switches

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Properties of an Ideal Switch

1. $V_{on} = 0$ and $V_{switch} = 0$

In the ON state of an ideal switch, there is no resistance. This means the switch creates a completely unobstructed path for current flow. As a result, it has no impact on the current, allowing it to pass through as if the switch were a perfect wire. Additionally, the switch acts as an ideal conductor, meaning it does not cause any energy loss in the circuit. Indication of this is evident in the fact that there is no voltage drop across the switch, ensuring that all supplied voltage is available to the connected load. Power dissipation is given by $P = VI$ or $P = I^2R$. Since both the resistance, R_{on} , and voltage drop, V_{drop} , are zero, the power dissipated by the switch is also zero. This makes an ideal switch 100% efficient, meaning no electrical energy is lost as heat, and all power is effectively transferred to the intended circuit components.

2. $R_{off} = \infty$ and $I_{off} = 0$

When an ideal switch is OFF, the resistance increases to infinity. This occurs because an ideal switch is assumed to have perfect insulating properties, meaning there is no conductive path between its terminals. As a result, it acts like a completely open circuit. This ensures that no electrical energy can pass through. Additionally, no current flows through the switch in the OFF state. Since current flow requires a complete circuit and the switch creates a break in the path, the current remains at 0, $I_{off} = 0$. This characteristic is critical for fully isolating different sections of a circuit, preventing unintended current leakage.

3. V_1 and V_2 value is unlimited

There are two terminals in a switch, V_1 and V_2 , which represent the voltages at each end of the switch. In an ideal switch, these voltage values can be infinitely large, either positively or negatively. This means the switch has no maximum voltage limitation. When the switch is ON, it acts like a perfect conductor, meaning there is no resistance, and no voltage drop across the terminals ($V_{switch} = 0$). Regardless of the values of V_1 and V_2 , the switch ensures that the voltage at both terminals remains equivalent, essentially shorting them together without introducing any losses or resistance. When the switch is OFF, it behaves as a perfect insulator, meaning no current can flow between V_1 and V_2 . An ideal switch can withstand any voltage difference between its terminals without failure. This means it can isolate parts of a circuit perfectly, regardless of the applied voltage.

4. Bidirectional

In an ideal switch, current can flow both forward and backward through the terminals. This means the switch is bidirectional and does not impose any restrictions on the direction of current flow. When the switch is ON, it acts as a perfect conductor ($R = 0$), ensuring current flows freely between terminals regardless of directions. There is no voltage drop across the switch ($V_{switch} = 0$), allowing the full voltage to reach the load without loss. When the switch is OFF, it acts as a perfect insulator ($R = \infty$), ensuring no current flows in either direction. This creates complete isolation between the two terminals, regardless of direction and voltage applied.

Non-Idealities of a Real Switch

1. Value of R_{on} measures non-ideality in switch when on

In an ideal switch, R_{on} would be 0. This means that there is no voltage drop and no power dissipation across the switch. However, in a non-ideal switch R_{on} is not 0. When the switch is in conduction, it will have a small, finite value that introduces a voltage drop and power loss. The higher the value of R_{on} is, the larger the voltage drop across the switch and the power dissipated by the switch. Both factors result in a reduced efficiency in applications. To measure the value of R_{on} , a multimeter can be used. To do so, the switch must be on, and the multimeter should be connected to both terminals of the switch. Ideally, the multimeter should read a low resistance across the switch.

2. Value of I_{off} measures non-ideality in switch when off

For an ideal switch, when it is off, the resistance is infinite ($R_{on} = \infty$). This means that no current leaks ($I_{off} = 0$). However, in a real switch, there will be some small leakage current even if the switch is off. This is because the switch is not a perfect insulator. This causes a loss of power when the switch is off. To measure the leakage current (I_{off}) connect an ammeter in series with the switch and apply a known DC voltage. Then, turn the switch off and measure the current reading. It will typically be small ($\mu A - nA$).

3. $V_{min} < V_1$ and $V_2 < V_{max}$

An ideal switch can handle any voltage at its terminals. However, non-ideal switches have maximum voltage ratings. Surpassing these values can lead to them breaking down. As well, they have minimum voltage ratings. Going below this value can make operating the switch unreliable. Although V_{max} and V_{min} can be calculated, they should be listed in the component's datasheet. V_{max} values can be found in the datasheet under absolute maximum ratings or electrical characteristics. Although, V_{min} values might not be explicitly listed in the datasheet, they can be inferred from the threshold voltage value.

4. Same value of R_{on} for $V_1 < V_2$ and $V_1 > V_2$

An ideal switch is bidirectional, meaning R_{on} is constant regardless of current direction. However, for a real switch the R_{on} varies based on the direction of current flow. This occurs due to the internal structure of the switch. To see the magnitude of R_{on} based on direction, measure R_{on} in both directions ($V_1 > V_2$ and $V_1 < V_2$). This can be done by connecting a current source with known value to the switch, then measuring the voltage across the switch using a voltmeter. Through Ohm's law ($R_{on} = \frac{V_{switch}}{I}$), the value of R_{on} can be calculated for.

References

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[2] O. A. Ahmed, "Power semiconductor devices," University of Technology, <https://odayahmeduot.wordpress.com/wp-content/uploads/2015/11/lecture-02.pdf> (accessed Feb. 15, 2025).